

# CI/CD Pipeline

## Overview

This CI/CD pipeline uses GitHub for source control, GitHub Actions for CI validation, AWS CodeBuild for building Docker images, Amazon ECR for storing images, and AWS Lambda for deployment.

## Workflow Process:

1. Developer creates a feature branch.
2. Code is pushed to the feature branch.
3. GitHub Actions runs CI validation.
4. Pull request is created to main branch.
5. After review, PR is merged.
6. CodeBuild builds Docker image.
7. Image is pushed to ECR.
8. Lambda function is updated with latest image.

## GitHub Actions:

CI runs on pull requests to main and pushes to feature branches. It checks out the repository, prints commit information and confirms validation success.

## Build and Deployment (CodeBuild):

CodeBuild logs into Amazon ECR, builds the Docker image using the Dockerfile, tags the image with commit SHA, pushes the image to ECR and updates the Lambda function.

## Project Structure:

Dockerfile – Defines Lambda Python runtime and container setup.  
buildspec.yml – Defines build and deployment steps.  
lambda\_function.py – Lambda handler.  
service.py – Business logic.  
requirements.txt – Python dependencies.

## Result:

After successful deployment, Lambda runs the latest container image and returns the updated response.